

Endogenous Heterogeneous Gender Norms and Female Labor Supply in an Intra-Household Bargaining Model

ODD-style model description of `gender_model_public_final.nlogo`

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1 Overview

1.1 Purpose

The model explains female (and male) labor supply as the joint outcome of (a) intra-household Nash bargaining over market work and an income transfer, and (b) endogenous, heterogeneous gender norms transmitted through same-sex social networks.

Heterogeneity in wages, preferences for private vs. public consumption, and conformism, interacted with network structure, generates dispersion and persistence in working time and transfers. Comparative experiments vary the female wage, preference distributions, conformism, utility specification, and network topology (random, Watts–Strogatz, preferential attachment, similarity-based, homophily, none, homogenous mixing).

The implementation analysed here is `gender_model_public_final.nlogo`, procedures `setup`, `go`, `set-theta`, `choose-bundle`, `calculate-utility`, `calculate-payoff`, `update-statistics`.

1.2 Entities, state variables, and scales

There are two agent breeds plus two link/collection entities:

Entity	State variables	Notes
women men (turtle breeds)	current-working-time in [0,1] old-working-time current-transfer-to-woman theta in [-1,1] old-transfer-to-woman spouse (partner pointer) wage >= 0 preference-private in [0.01,0.99] conformism >= 0 current-utility, memory-change, number	One woman + one man = one household. number only used in import-csv mode. memory-change is solver scratch state.
Household (implicit)	(woman, man, theta) triple; theta>0 = transfer man→woman	No separate NetLogo agent; bargaining unit in set-theta. Outside option defined at theta=0.

Entity	State variables	Notes
Social links (links breed)	Undirected same-sex ties	Single breed shared by both sexes but only linked within sex, except via spouse-of-neighbour perception.
Observer (globals)	global-mean-working-time-men/ women, global-mean-transfer, history-*, average-mean-*, sub- group agentsets	Subgroup agentsets fixed at setup from initial medians; histories are 10-period moving windows.

Scales. No explicit space matters: layout-circle is visualisation only. Time is discrete ticks; one go = one round of household re-bargaining + norm updating. Experiments run 10–100 ticks. Labour and transfer are normalised fractions, not hours.

1.3 Process overview and scheduling

Phase	What happens
setup	Clear, seed RNG if fixed-rs. Generate same-sex networks (for random-network, watts-strogatz, preferential-attachment). Create agents, pair spouses, draw wage / preference-private / conformism / theta from truncated normals around slider means. Optionally override one specific-couple or overwrite from CSV. Fix subgroup agentsets. Optionally build similarity networks (preference-private, wage, conformism, homophily). reset-ticks.
go, set-theta	For each woman in sequence: (1) compute outside options via choose-bundle(self, spouse, 0); (2) hill-climb theta up then down in steps of 0.001 using calculate-payoff; (3) commit best-theta and the associated choose-bundle labour pair to both spouses.
go, choose-bundle	Iterated best response: woman and man alternately hill-climb own h by current-delta = 0.00001 while partner h is held fixed, until neither changes. Evaluates calculate-utility.
go, update-statistics	Copy current-* to old-*. Recompute global means, 10-period histories and moving averages.
go, tick + I/O	Advance clock; optionally append per-turtle row via print-agent-positions.

Ordering is sequential over households within a tick. Norm perception uses lagged old-* values, so intra-tick ordering does not directly enter norms, but choose-bundle reads partner current-working-time while it is being mutated.

2 Design concepts

Concept	Realisation in this model
Basic principle	Collective household model with Nash bargaining + conformity-augmented CES/additive utility + social norms. Norms are local means, hence heterogeneous and endogenous.
Emergence	Gender gaps in hours, transfer distributions, and norm clusters emerge from bargaining + network topology; not imposed beyond initial means norm-paid-time-male/female.
Adaptation	Households adapt h_w, h_m, theta by myopic hill-climbing. No forward-looking search, no link rewiring after setup.
Objectives	Maximise own calculate-utility (labour stage) and Nash product ($U_m - U_{out_m}$) ($U_w - U_{out_w}$) (transfer stage). Individual rationality required, else payoff -1.

Concept	Realisation in this model
Learning / Prediction	None. One-period lagged imitation via norm penalty only. No expectation formation.
Sensing	Agents sense: own + spouse wage/preference/conformism; partner current h; mean old-h and old-theta of same-sex link-neighbors; mean old-h of spouses-of-neighbors. If isolated: own lag, or global mean under homogenous mixing.
Interaction	Direct: intra-household bargaining. Indirect: inter-household via norm penalty $\exp(\text{norm-parameter})$. No market clearing.
Stochasticity	Initial draws (truncated normals), spouse matching one-of, network generators, <code>rnd:weighted-n-of noise (N(0,0.001))</code> . Reproducible via <code>fixed-rs + random-seed-fixed</code> .
Collectives	Household (bargaining unit); same-sex neighbourhood (reference group); median-split subgroups for observation only.
Heterogeneity	Continuous heterogeneity in wage, preference-private, conformism; structural heterogeneity via degree/position.
Observation	Means/medians/SD/p25/p75 of utility, hours, transfers; subgroup means by wage/preference; moving averages; per-agent CSV trace; time-series plot + histograms.

3 Details

3.1 Initialization (setup, set-initials-*)

1. `ca`; set `current-delta = 0.00001`, clear histories.
2. If `fixed-rs`, `random-seed random-seed-fixed`.
3. Build `random-network / watts-strogatz / preferential-attachment` networks via `nw` extension **before** agents get traits; other modes create unlinked turtles first.
4. `layout-circle` (visual only); re-seed if `fixed-rs` so trait distributions are comparable across topologies.
5. Pair each woman with one-of men with [`spouse = 0`]; reciprocal spouse pointers.
6. Women: `h = norm-paid-time-female/100`, `theta ~ N(initial-transfer, initial-transfer * std-dev-values)` clipped to `[0,1]`, `wage ~ N(wage-female, mean(wage-male,wage-female)*std-dev-values)` clipped at 0, `conformism ~ N(conformism-female, ...)` clipped at 0, `preference-private ~ N(preference-private-mean-female, ...)` clipped to `[0.01,0.99]`. Men symmetric with male sliders and `norm-paid-time-male/100`; man inherits initial `theta` from spouse.
7. Optional `specific-couple` override; median splits for high/low public-goods households and above/below median wages (static snapshots).
8. Optional similarity/homophily network construction; optional CSV overwrite (`initialTransfer.csv`, `moderatePreferences.csv`, `highConformity.csv`, `moderateConformity.csv` — note: preferences file is read twice, second read overwrites the first).
9. `reset-ticks`; optionally open `agent_position_<exp>_<run>.csv`.

Key defaults in the Interface tab: `wage-female=0.75`, `wage-male=1.0`, `norm-paid-time-male=80`, `norm-paid-time-female=20`, `number-agents-each-type=20` (experiments use 1 or 401), `preference-private-mean-{male,female}=0.5`, `initial-transfer=0.1`, `std-dev-values=0.2`, `conformism-male=3.0`, `conformism-female=0.0`, `utility-function=CES` with `CES_beta=0.5`, `weight-working-time-self/partner/transfer=1`.

3.2 Input data

No time-series input. `import-csv=true` replaces random traits with files listed above. CSV handling assumes row order matches household number and provides no existence/length checks. All BehaviorSpace experiments shipped with the file use `import-csv=false`.

3.3 Submodels

3.3.1 Norm perception (calculate-utility, lines 561–586)

Let h_i be own hours, h_j spouse hours, θ transfer-to-woman.

N_i^h = mean old-h of link-neighbors of i

N_i^θ = mean old-theta of link-neighbors of i

$N_i^{h_j}$ = mean old-h of spouses-of-neighbors

$$\text{norm-penalty}_i = -c_i \cdot \left(\omega_s (h_i - N_i^h)^2 + \omega_\theta (\theta - N_i^\theta)^2 + \omega_p (h_j - N_i^{h_j})^2 \right)$$

c_i is own conformism; ω are the three weight-* sliders. Isolated agents use own lag; under homogenous mixing they use the global means instead.

3.3.2 Material utility and conformity multiplier

Define private consumption x_i and domestic good $Q = 2 - h_i - h_j$:

- If recipient (relevant-transfer < 0, sign-flipped for women): $x_i = h_i w_i + |\theta| w_j h_j$.
- If donor: $x_i = h_i w_i (1 - \theta)$.
- If $x_i \leq 0$ or $h > 1$: utility = -200 (infeasibility penalty).

Four utility-function options (with α_i = preference-private, β = CES_beta):

- additive: $(\alpha_i \sqrt{x_i} + (1 - \alpha_i) \sqrt{Q}) \cdot \exp(\text{norm-penalty}_i)$.
- CES: $(\alpha_i x_i^\beta + (1 - \alpha_i) Q^\beta)^{\frac{1}{\beta}} \cdot \exp(\text{norm-penalty}_i)$.
- multiplicative: $\sqrt{x_i} \sqrt{Q} \cdot \exp(\text{norm-penalty}_i)$.
- multiplicative including weights: intended $x_i^{\alpha_i} Q^{1-\alpha_i} \cdot \exp(\dots)$; code as shipped computes $((x_i^{\alpha_i} \cdot Q))^{1-\alpha_i}$ – see code critique.

3.3.3 Labour best response (choose-bundle)

Holding θ fixed, each partner steps h_i by $\pm$ current-delta while it strictly improves own utility and stays in $[0, 1]$. Partners alternate until a full round yields zero change. memory-change (last 10 steps) and sum-memory-change guard against 2-cycles. No step-size annealing is active in the shipped version (the current-delta / 2 block is commented out).

3.3.4 Transfer bargaining (set-theta, calculate-payoff)

Outside options U_w^0, U_m^0 are the choose-bundle utilities at $\theta = 0$. Candidate θ values are tested in two one-sided hill-climbs from the current θ with step 0.001 over $[-1, 1]$, stopping at the first decrease per side:

$$\text{payoff}(\theta) = \begin{cases} (U_{m(\theta)} - U_m^0)(U_{w(\theta)} - U_w^0) & \text{if both } \geq 0 \\ -1 & \text{otherwise} \end{cases}$$

The θ with the highest payoff and its choose-bundle hours are committed. Note: best-payoff is initialised to 0 and best-theta to 0, so the status quo is never evaluated and a no-improvement search defaults to theta=0.

3.3.5 Network formation (generate-network, generate-homophilic-network)

Similarity networks connect each turtle to preferential-attachment-min-degree others sampled by rnd:weighted-n-of with weight $1/|x_{\text{myself}} - x_{\text{other}} - N(0, 0.001)|$ for $x \in \{\text{preference-private, wage, conformism}\}$. Homophily uses $1/(|\delta \text{ conformism}| + |\delta \text{ wage}| + |\delta \text{ preference}|)$. Ties are created once at setup and never updated.

3.3.6 Statistics (update-statistics)

Copies current-* to old-*, recomputes global means, pushes them onto 10-element histories, and reports their moving averages average-mean-working-time-{women, men}. global-mean-transfer is the contemporaneous mean over women.

4 Parameters, experiments, and output

Parameter	Default	Meaning
number-agents-each-type	20	Households; experiments use 1 (analytic) or 401.
wage-{female,male}	0.75 / 1.0	Mean wage offers; dispersion via std-dev-values.
preference-private-mean-{female,male}	0.5 / 0.5	Mean weight on private vs. domestic good.
conformism-{female,male}	0.0 / 3.0	Mean norm sensitivity.
std-dev-values	0.2	Relative SD scaler for all traits.
norm-paid-time-{female,male}	20 / 80	Initial hours in percent.
initial-transfer	0.1	Mean initial θ .
utility-function, CES_beta	CES, 0.5	Aggregator and substitution parameter.
weight-working-time-self, weight-transfer, weight-working-time-partner	1 / 1 / 1	$\omega_s, \omega_\theta, \omega_p$.
network-structure	—	random-network, watts-strogatz, preferential-attachment, preference-private, wage, conformism, homophily, none, homogenous mixing.
random-network-prob, watts-strogatz-neighbors/rewiring, preferential-attachment-min-degree	0.01, 2 / 0.1, 2	Topology hyperparameters.
fixed-rs, random-seed-fixed	on/off, 1	Reproducibility switch and seed.
specific-couple, *-in-question	off	Single-household probe overrides.
import-csv, output-locations	off	CSV trait import; per-agent trace export.

Shipped BehaviorSpace: experiment_1 (single couple, wage/preference/ β sweep); experiment_2–9 (401 households $\times$ 1000 seeds $\times$ wage-female {0.75, 1.0, 1.25} $\times$ symmetric vs.

asymmetric preferences, one network each); different_conformism (conformism-male 0–5). Metrics: mean/median/SD/p25/p75 of utility, hours, transfers, plus subgroup means. runMetricsEveryStep=false — end-of-run snapshots only.

5 Known implementation quirks (for replicators)

1. Specific-woman conformism override uses the male probe value.
2. Preferences CSV block is duplicated.
3. set-theta never evaluates staying put; defaults to 0 on failure.
4. multiplicative including weights parenthesisation differs from the Cobb–Douglas intent.
5. precision $\times 8$ in calculate-payoff creates plateaus at the $1e-5$ labour step.
6. calc-pct indexing is only safe for interior percentiles (used as 25/75).

7. Subgroup agentsets and Info-tab documentation are static/stub as shipped.

ODD protocol after Grimm et al. (2006, 2010, 2020). This description was reverse-engineered from code; ambiguous points are noted above rather than resolved. Source: `gender_model_public_final.nlogo` + Interface/BehaviorSpace sections.